A Shadow-Robust Pavement Damage Detection Framework Based on RACycle-GAN and DDE-YOLOv8

Weiyi Kong, Yicheng Li, Jing Hu, Sang Luo, and Jiangqiu Rui

***Abstract*— Pavement construction worldwide has expanded significantly, making maintenance of pavement infrastructure a highly challenging task. Traditional manual inspection methods are inefficient and costly, while deep learning-based object detection algorithms often underperform in complex scenarios such as shadowed pavement surfaces. To address these issues, by combining a shadow removal model (Residual Attention Cycle GAN, RACycle-GAN) and Damage Detection Enhanced YOLOv8 (DDE-YOLOv8), this paper proposes a shadow-robust pavement damage detection framework. By integrating the Residual Encoder-Decoder Block, two-branch spatial attention, and cycle consistency into the original DCGAN architecture, the proposed RACycle-GAN significantly enhances shadow removal performance, fundamentally solving the issue of detection under shadow interference. Additionally, DDE-YOLOv8 is introduced to further enhance the damage detection performance. Experimental results demonstrate that RACycle-GAN greatly improves detection performance, and DDE-YOLOv8 outperforms the original version. The combination of RACycle-GAN and DDE-YOLOv8 achieves a leading performance in pavement damage detection, demonstrating the effectiveness of the shadow-robust damage detection method in this paper.**

***Index Terms*— Image translation, generative adversarial network, damage detection, YOLOv8, pavement inspection.**

I. Introduction

As a fundamental element of transportation infrastructure, pavements have been under increasingly extensive construction in many countries. Pavements are subjected to heavy traffic and sustained freight loads, making them highly vulnerable to various types of damage and deterioration. Consequently, the efficiency and precision of pavement damage detection are crucial for effective maintenance. A central focus of research in intelligent pavement maintenance is a reliable solution to timely, efficient, and accurate damage detection [3], [40].

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Currently, pavement maintenance work still heavily relies on manual inspection which is highly inefficient and costly, and difficult to standardize the criteria for identifying damage. Additionally, manual data recording and analysis are time-consuming, with extremely limited daily inspection distance [32]. Subsequently, many researchers began utilizing object detection methods based on traditional classification algorithms [11], [26] and grayscale threshold segmentation algorithms [8], [33] to achieve automated damage detection. However, the methods mentioned above are only effective for images where there is a significant difference in grayscale values between damages and the background. Since pavement damage images obtained in practice often contain complex interference factors and irregular grayscale distributions, accurate damage detection becomes extremely challenging. Deep learning methods, noted for excellent performance and generalization capabilities, have gained widespread application in damage detection in recent years [23], [52], [53], [54], [55]. However, shadows cast on pavement surfaces can introduce noise into the images, interfering with the features of pavement damage near the target regions. These shadows appear as areas with varying brightness, diminishing the feature extraction performance of deep learning models. Consequently, shadow removal has become a critical issue in enhancing the performance of damage detection.

Traditional shadow removal methods can generally be categorized into three approaches: image preprocessing [18], [48], [56], multispectral imaging [30], [37], and data augmentation [1], [7], [50]. However, image preprocessing tends to suffer from issues such as color distortion, image artifacts, and uneven brightness. Nonetheless, despite the advantage of multiwavelength data, multispectral imaging typically imposes stringent requirements on imaging equipment. Additionally, excessive data augmentation may lead to the loss or distortion of the original data information, resulting in features being unrealistic or inaccurate. In recent years, image translation algorithm has become a new research direction of shadow removal by using neural networks to establish the mapping relationship between shadowed images and shadow-free images. However, typical image translation algorithms may have problems such as high computational cost, mode collapse, insufficient use of image information, and difficulty in focusing on the shadow feature regions, which are not suitable for pavement inspection scenarios. Aiming at solving the problems above, this paper proposed a novel algorithm:

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RACycle-GAN. Specifically, a two-branch attention mechanism is introduced to guide RACycle-GAN focusing more on the high-level features of shadow regions. Subsequently, the Residual Encoder-Decoder Block is designed to retain more high-level features of shadow and maintain efficient use of information flow. Additionally, we employ cycle consistency loss to enable the generator to completely use input image information and improve stability.

In pavement damage detection, different types of pavement damage can vary significantly in scale within images, exhibiting a wide range of shape features. This diversity poses considerable challenges for detection algorithms. Therefore, apart from shadow interference, the complexity and diversity of features and textures of damages present significant challenges to detection algorithms. Furthermore, DDE-YOLOv8 is introduced to further enhance damage detection performance. By employing deformable convolution blocks and CBAM, DDE-YOLOv8 enhances the detection capability of multi-scale damages.

The contributions of this paper are summarized below.

1) In this paper, we propose a novel RACycle-GAN by introducing the Residual Encoder-Decoder Block, two-branch spatial attention mechanism, and cycle consistency to enhance the performance of shadow removal.
2) With the assistance of deformable convolution blocks and CBAM, DDE-YOLOv8 acquires the great ability of multi-scale damage detection.
3) The comprehensive experimental results demonstrate the advantage of the proposed RACycle-GAN over compared algorithms.
4) DDE-YOLOv8 surpasses the compared damage detection algorithms. Moreover, employing RACycle-GAN significantly improves the performance of pavement damage detection algorithms.

II. Related Works

A. Image Translation

The shadow removal method based on image translation essentially uses the neural network to generate the target style or target domain images from given input images. Early popular image translation algorithms include autoencoder [17]. Fan et al. [10] proposed an unsupervised, end-to-end autoencoder model for learning cross-domain mappings between shadow and shadow-free images, while studies have shown that autoencoder is prone to artifacts when removing shadows [31]. By introducing probabilistic modeling, the Variational Autoencoder (VAE) [21] was then developed on the basis of the autoencoder [17]. Bulatov et al. [4] used VAE to separate and reduce the dimension of shadow regions in remote sensing data. However, the images generated by VAE may suffer from low quality and poor sharpness [6].

Diffusion model [39] has made notable advancements in the field of generative models in recent years, whose potential applications in shadow removal have also gained attention in recent research. Reference [29] utilized the diffusion model to infer shadow-free scenes from shadow images, demonstrating excellent instance-level shadow removal performance. Guo et al. [16] improved diffusion model by integrating images and degenerate priors, resulting in more accurate and robust shadow-removed images. Luo et al. [28] proposed a global guided diffusion model for high-quality shadow removal, which solved the situation that diffusion model ignored the global information and led to inconsistent illumination. Conversely, the diffusion model is computationally expensive. It requires a gradual de-noising process, resulting in long processing times for each image, leading to poor real-time performance.

Generative adversarial network (GAN) [15] and its variants have been one of the most popular methods in the generative task, and are currently the mainstream methods in image translation. Zhang et al. [57] introduced context and lightness aware Generative Adversarial Network (CLA-GAN) framework for shadow removal. Zhang et al. [58] proposed a general and novel framework RIS-GAN which explores residual and illumination with GAN for shadow removal. Wang et al. [44] introduced a novel STacked Conditional GAN (ST-CGAN), consisting of two stacked CGANs which work together for end-to-end shadow removal. However, GAN-based models are prone to pattern collapse problems, resulting in insufficient use of the valid information of input images. In order to effectively solve the problem, CycleGAN introduced circular structure and circular consistency. Hu et al. [20] proposes Mask-ShadowGAN, automatically learning to generate shadow masks from input shadowed images, and then using masks to guide shadow generation through reformulated circular consistency constraints. This means that the method must train with shadow mask data, increasing the difficulty of establishing the dataset. In recent years, many studies have focused on integrating emerging neural network technologies into GAN-based shadow removal algorithms, such as ShadowGAN-Former [19] with self-attention mechanisms, an improved EnlightenGAN [41] with heuristic enhancement, and DDSR-Net [43] based on multi-scale attention.

Even though CycleGAN addresses some limitations of the original GAN, it still encounters two primary issues. First, pavement shadows present multi-scale features, which makes it difficult for CycleGAN to focus on the shadow feature regions. Second, the image information is easily lost when the generator generates shadow-removed images from high-level features. Aiming at addressing these problems, a Two-branch Spatial Attention and Residual Encoder-Decoder Block are introduced to RACycle-GAN.

B. Damage Detection

Pavement damage detection is a combination of location and classification in computer vision tasks, i.e., the algorithm locates the location of damages in the image and classifies them into a specific damage category. Currently, widely used object detection networks can be categorized into two-stage and one-stage architectures. Two-stage detection networks include models such as R-CNN [14], Fast R-CNN [13], and Faster R-CNN [36], which first generate region proposals and then classify these proposals to detect objects. In the early stages of research, two-stage object detection networks were

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TABLE I
DETECTION RESULT COMPARISON OF INTERFERENCE FACTORS

Metrics	Interference Factors			
	Shadows	Pavement Markings	Expansion Joints	Water Stains
Accuracy	**0.80**	0.90	0.98	0.92
Precision	**0.83**	0.91	0.98	0.96
Recall	**0.96**	1	1	0.96

utilized for pavement damage detection [2], [22], [34], [38], [45]. However, these networks typically demonstrate slower processing speeds compared to single-stage models due to the additional proposal generation. Therefore, the single-stage object detection algorithm represented by YOLO has become the mainstream of pavement damage detection.

Du et al. [9] contributed a large-scale dataset of pavement damage and used YOLOv3 to predict the location and category of possible pavement damage. By enhancing YOLOv5, Wan et al. [42] proposed a lightweight model for pavement damage identification. Wu et al. [47] proposed an efficient lightweight pavement damage detection algorithm for mobile terminal devices, YOLO-LWNet, which better realized the lightweight and precision requirements of detection for mobile terminal devices. However, YOLO abandons the operation of generating proposals, resulting in slightly lower detection accuracy, especially when dealing with small targets or complex backgrounds, which may not be as accurate as two-stage models. How to make YOLO better extract damage features is a significant research problem. Yao et al. [51] discussed the impact of different methods and positions to add attention modules into the YOLOv5 model. Xie et al. [49] introduced a pavement damage detection algorithm based on feature diffusion. Geng et al. [12] propose the Selective Dynamic Feature Compensation-YOLO (SDFC-YOLO) algorithm for pavement damage detection. Li et al. [25] integrate the simple attention mechanism into the backbone to improve the focus on crucial details. In this paper, by employing deformable convolution and CBAM, the ability of the proposed DDE-YOLOv8 to sense multi-scale damages has been improved.

III. METHODOLOGY

A. Analysis of Pavement Interference Factors

In pavement damage detection, the detection performance is often affected by interference from complex pavement environments. In the Pavement Damage Dataset (PDD) used in this paper, the primary interference factors include shadows, pavement markings, expansion joints, and water stains. To quantify the impact of these four interference factors on damage detection performance, an experiment is performed using the original YOLOv8 model trained on PDD. A dataset of 200 images mixed with interference factors and damages, with 50 images for each interference factor, is selected to evaluate the performance of the trained the original YOLOv8 model. The results are shown in Table I. Detailed explanations of the used metrics can be found in Section IV-C.

As indicated in Table I, under shadow interference, although recall is still maintained at 96.1%, precision drops to 83.3%. Moreover, the accuracy of the damage classification drops to 80%, demonstrating that shadows are the most significant interference factor impacting detection performance. Therefore, we propose an improved Shadow Removal algorithm based on Deep Convolutional Generative Adversarial Network (RACycle-GAN) to mitigate the shadow interference in damage detection, thereby enhancing the pavement damage detection performance.

B. RACycle-GAN

1) Architecture: The proposed RACycle-GAN is primarily based on the DCGAN [35] framework, which originates from GAN. GAN consists of two main components: a generator and a discriminator. Through continuous optimization of the dynamic adversarial process between these two components, GAN is able to approximate the distribution of real data and generate synthetic data. DCGAN improves upon traditional GAN by adopting a deep convolution network for both the generator and the discriminator, enhancing the adversarial training process, and leading to faster model convergence. However, DCGAN has the following three shortcomings: the image information is easily lost when the generator generates shadow-removed images from high-level features; being difficult to focus on the shadow feature regions; being easy to fall into mode collapse and cannot fully learn the mapping relationship between shadowed image and shadow-free image. Building upon the DCGAN framework, this paper introduces three key improvements to solve the shortcomings above:

1) Introducing Residual Encoder-Decoder Block to help retain more high-level features of shadow and maintain efficient use of information flow.
2) Embedding Two-branch Spatial Attention within both the generator and discriminator to guide the focus of the network on regions that better match the features of shadow.
3) Incorporating a cycle consistency loss to prevent mode collapse, ensuring that the generated data remains diverse and stable during the adversarial process.

The proposed RACycle-GAN is consists of two generators ($G_{s\to sr}$, $G_{sr\to s}$) and a discriminator D, as illustrated in Figure 1. First, a shadowed image I_s is fed into $G_{s\to sr}$, with high-level features being extracted by the Residual Encoder Block. After the feature passes through the spatial attention mechanism, a Residual Decoder Block is employed to restore the generated shadow-removed image I'_{sr}. We then randomly select an image between I'_{sr} and a real shadow-free image I_{sf} corresponding to I_s and feed it into the discriminator to make a judgment whether it is real or generated. Meanwhile, we design another generator $G_{sr\to s}$ to reconstruct the original image from I'_{sr}, denoted as I'_s. The cycle consistency loss is calculated between I_s and I'_s, and the parameters of both generators and discriminator are optimized end-to-end. After abundant epochs of adversarial iteration, the discriminator becomes increasingly accurate at making correct judgments, which in turn promotes the generator to produce progressively more realistic shadow-free images.

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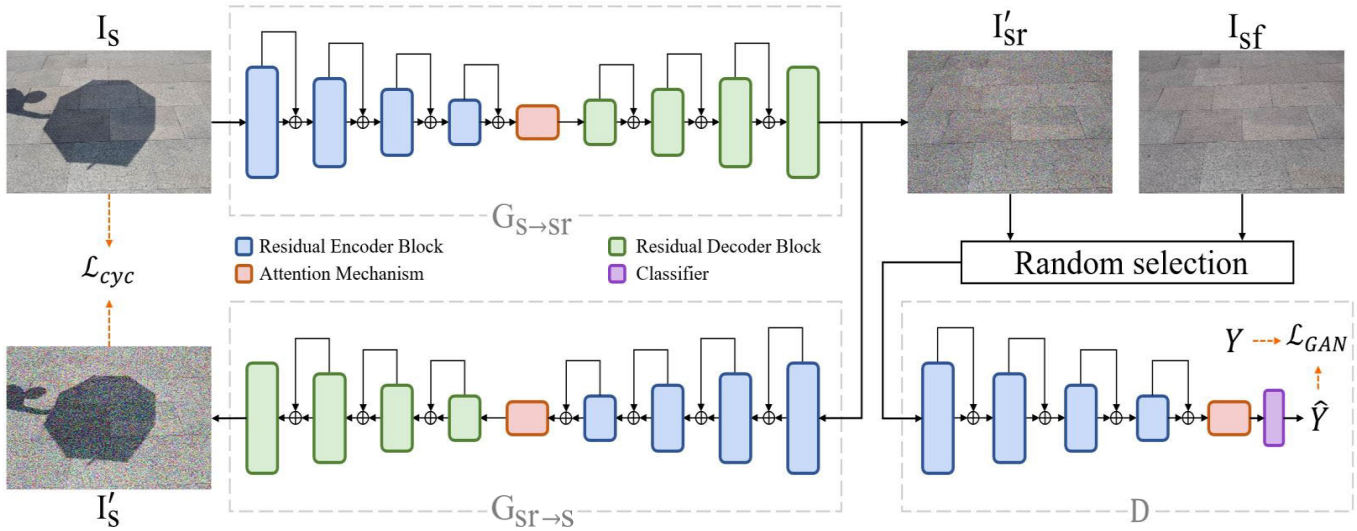

Fig. 1. Architecture of RACycle-GAN.

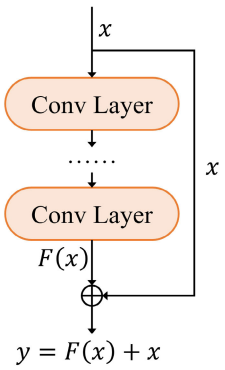

Fig. 2. Illustration of Residual Connection.

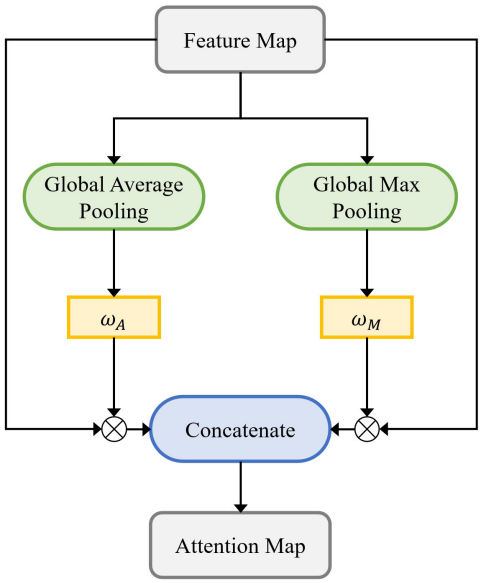

Fig. 3. Architecture of Two-branch Spatial Attention.

2) Residual Encoder-Decoder Block: Residual connection is a technique used in deep neural networks to combat the problem of vanishing gradients and improve the training of very deep networks. It was proposed by the ResNet [60] architecture.

In RACycle-GAN, the Residual Encoder-Decoder Block is employed in the generators and the discriminator to help retain more high-level features of shadow and maintain efficient use of information flow. For the input to a layer x, the residual connection allows it to bypass one or more layers, as shown in Figure 2. After producing a transformation $F(x)$, x is added directly to $F(x)$, which makes the output of these bypassed layers as $y = F(x) + x$, allowing information to flow directly to deeper layers without being affected by the transformations of intermediate layers.

3) Two-Branch Spatial Attention: RACycle-GAN incorporates Two-branch Spatial Attention in both generators and the discriminator to ignore irrelevant information, better learn shadow features, and enhance the focus on shadow regions.

The architecture of the Two-branch Spatial Attention is illustrated in Figure 3. The feature map $X \in \mathbb{R}^{H\times W\times C}$ extracted by the Residual Encoder Block is fed into the Two-branch Spatial Attention. Through global average pooling ($GAP(\cdot)$) and global max pooling ($GMP(\cdot)$), two learnable weight parameters ω_A and ω_M are generated:

$$\omega_A = GAP(X) = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} x_{ij} \tag{1}$$

$$\omega_M = GMP(X) = max(\sum_{i=1}^{H} \sum_{j=1}^{W} x_{ij}) \tag{2}$$

ω_A and ω_M determine the importance of the corresponding features for each channel to emphasize important features while downplaying less relevant features. Subsequently, ω_A and ω_M are separately multiplied by X spatial-wisely. After feature concatenation, the weighted attention map is passed to subsequent network layers (which refer to the decoder in the generator and the classifier in the discriminator). This process enhances the focus on more relevant target features, thereby

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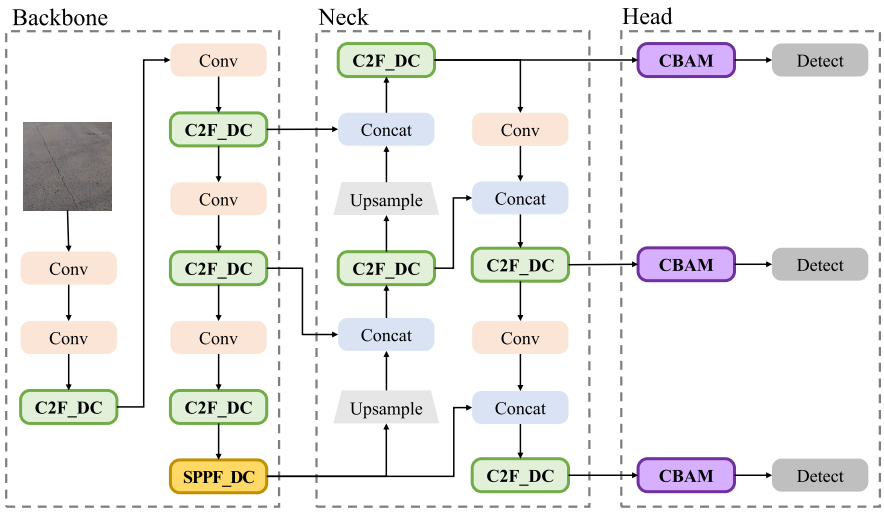

Fig. 4. Architecture of DDE-YOLOv8.

improving the overall performance of the model in generating shadow-free images and distinguishing between shadow-free and shadow-removed images.

4) Cycle Consistency: Mode Collapse refers to the generator collapsing to a few modes of the data distribution, leading to poor very limited outputs and a lack of diversity in the generated samples. To prevent mode collapse and enable the generators more efficiently learn the mapping relationship between shadowed images and shadow-free images, cycle consistency is incorporated into RACycle-GAN. Specifically, for an input shadowed image I_s, a shadow-removed image I'_{sr} is generated by $G_{s\to sr}$, which is then fed back into $G_{sr\to s}$ to reconstruct a shadowed image I'_s. A higher similarity of I'_s and I_s refers to a better cycle consistency of the network. The cycle consistency ensures that every image mapped from domain A (shadowed) to domain B (shadow-free) can be mapped back to its original form, therefore enhancing the ability of the generator to effectively generate shadow-removed images, ensuring that the model remains robust and capable of producing diverse outputs.

5) Loss Function: The loss function of RACycle-GAN consists of the following two components: adversarial loss $\mathcal{L}_{GAN}$ and cycle consistency loss $\mathcal{L}_{cyc}$.

The adversarial loss is a loss function utilized by $G_{s\to sr}$. It is designed to enable the generator to generate realistic shadow-removed images and to enable the discriminator to distinguish between shadow-free and shadow-removed images as precisely as possible. $\mathcal{L}_{GAN}$ is calculated as follows:

$$\begin{aligned}&\min_G \max_D \mathcal{L}_{GAN}(D, G_{s\to sr})\\&\quad = \mathbb{E}_{sf\sim p(sf)}[log D(I_{sf})] + \mathbb{E}_{s\sim p(s)}[log(1 - D(I'_{sr})]\end{aligned} \tag{3}$$

where I_{sf} and $I'_{sr} = G_{s\to sr}(I_s)$ represent shadow-free images and shadow-removed images respectively. $D(\cdot)$ denotes the judgment of the discriminator on the input images; $E(\cdot)$ represents the mathematical expectation; $\mathbb{E}_{sr\sim p(sr)}[\cdot]$ refers to the expectation of I_{sr} where I_{sr} follows the distribution $p(sr)$.

After $G_{s\to sr}$ translates I_s into I'_{sr}, another generator $G_{sr\to s}$ translates I'_{sr} into regenerated shadowed image I'_s, where cycle consistency loss $\mathcal{L}_{cyc}$ is calculated as:

$$\mathcal{L}_{cyc}(G_{s\to sr}, G_{sr\to s}) = \mathbb{E}_{real_s\sim p(s)}\left[\|I'_s - I_s\|_1\right] \tag{4}$$

where $\|I'_s - I_s\|_1$ represents the L1 norm. The generator $G_{sr\to s}$ depends $\mathcal{L}_{cyc}$ to optimize parameters.

With summing $\mathcal{L}_{GAN}$ and $\mathcal{L}_{cyc}$ with a hyperparameter λ ($\lambda = 10$ [59]) to control the importance of $\mathcal{L}_{cyc}$, the total loss function of RACycle-GAN is calculated as:

$$\begin{aligned}&\mathcal{L}(G_{s\to sr}, G_{sr\to s}, D)\\&\quad = \mathcal{L}_{GAN}(D, G_{s\to sr}) + \lambda\mathcal{L}_{cyc}(G_{s\to sr}, G_{sr\to s})(I'_{sr})]\end{aligned} \tag{5}$$

C. Damage Detection Based on DDE-YOLOv8

1) Architecture: YOLOv8 is a single-stage object detection algorithm that makes predictions based on the entire image. An improved pavement damage detection algorithm based on the YOLOv8 network architecture is proposed, seen in Figure 4. First, the C2f module in the backbone feature extraction network is optimized by replacing the original convolutional modules with deformable convolution DCNv3 module [24]. Additionally, 1×1 convolutions in the feature pyramid module are substituted with 3×3 DCNv3 modules, enhancing the awareness of multi-scale damage features. This adjustment enables the network to better aggregate image features of varying scales and improves the detection accuracy of pavement damage targets with significant scale differences. Furthermore, the top detection network incorporates the Convolutional Block Attention Module (CBAM), which utilizes a channel attention mechanism to assess the importance of different channel features. This addition further refines and emphasizes key features within the target regions, enhancing the overall detection performance. The loss function of DDE-YOLOv8 consists of the following parts: classification loss (cross-entropy loss) and bounding box regression loss (distribution focal loss and complete intersection over union loss).

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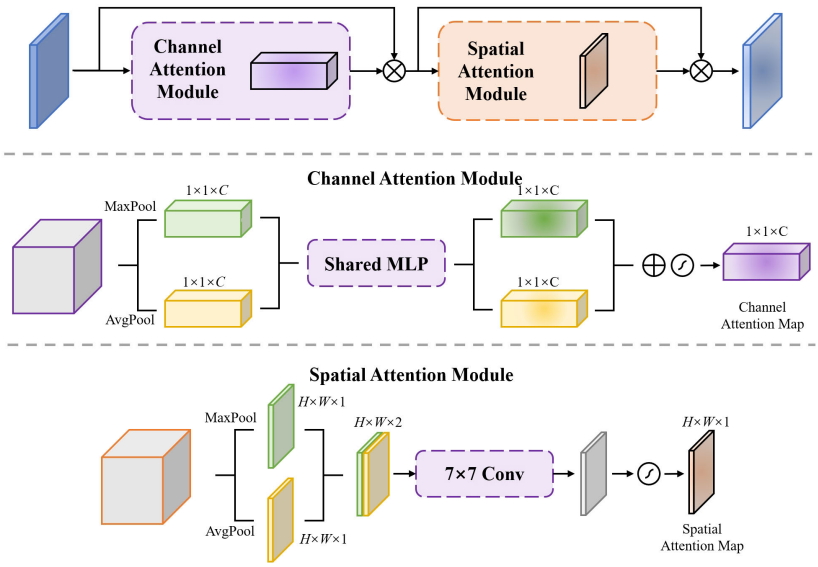

Fig. 5. The architecture of CBAM.

2) Deformable Convolution: Deformable convolution introduces learnable offsets, allowing the convolution kernel to dynamically adjust the sampling points at each position based on the content of the input feature map, illustrated as Figure 6. This enables the network to extract features more effectively. The formula for deformable convolution can be expressed as follows:

$$y(p_0) = \sum_{p_n \in R} w(p_0) x(p_0 + p_n + \Delta p_n) \tag{6}$$

where $y(p_0)$ is the output at location p_0. R represents the regular convolution grid (e.g., a 3×3 window). p_n are the predefined positions in the receptive field. $w(p_n)$ are the weights of the convolution kernel at position p_n. $x(p_0 + p_n + \Delta p_n)$ is the input feature at the deformed location $p_0 + p_n + \Delta p_n$, where Δp_n represents the learnable offset.

3) CBAM: Woo et al. [46] proposed CBAM whose architecture is seen as Figure 5, which is a lightweight and effective attention mechanism designed to improve the performance of CNN by refining feature maps at both the channel and spatial levels.

First, the channel attention module is applied to refine the feature maps by focusing on the most informative channels. The channel attention map is calculated as below:

$$M_c(X) = \sigma\left(MLP(AvgPool(X)) + MLP(MaxPool(X))\right) \tag{7}$$

where $M_c(X)$ is the channel attention map and σ is the sigmoid activation function. Multiply X by $M_c(X)$, denoting the result as X_c. After that, the spatial attention module takes X_c as the input to calculate the spatial attention map $M_s(X_c)$:

$$M_s(X_c) = \sigma\left(f^{7\times7}(concat[AvgPool(X_c); MaxPool(X_c)])\right) \tag{8}$$

where $f^{7\times7}$ represents a convolution operation with a 7×7 kernel size. Finally, the output of CBAM is calculated as $Y = M_s(X_c) \times X_c$.

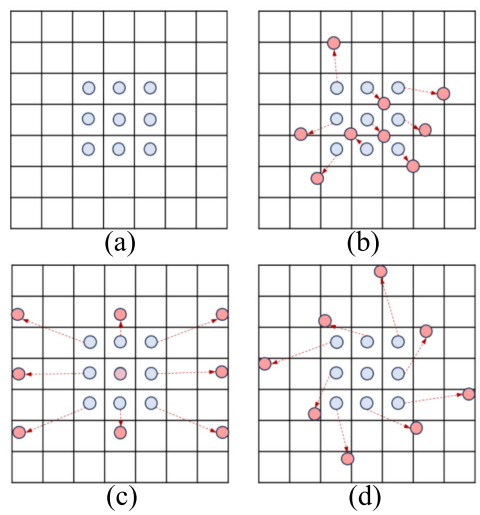

Fig. 6. Illustration of 3×3 standard and deformable convolutions. (a) standard convolution. (b) deformable convolution. (c)(d) special cases of deformable convolution.

IV. Experiments

A. Datasets

1) Shadow Removal Dataset: The Image Shadow Triplets Dataset (ISTD) [44] is a publicly available shadow dataset, as shown in Figure 7. It consists of 1870 pairs of shadow-related images (shadowed image, shadow mask image, shadow-free image) across 135 different scenes, designed for various shadow-related computer vision tasks such as shadow detection, generation, and removal.

The ISTD dataset is split into training and test sets in a 10:1 ratio. The training set contains 1,700 pairs of shadowed and shadow-free images along with pavement damage, while the test set comprises 170 pairs. The shadow removal algorithms discussed in this paper are trained and evaluated on this dataset.

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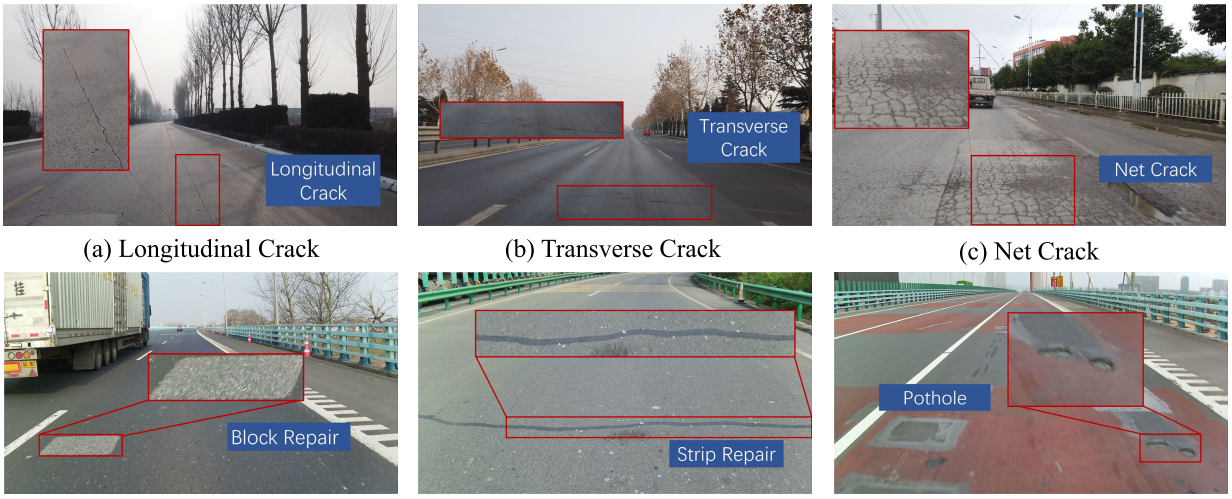
(a) Shadowed (b) Shadow-removed (c) Shadowed (d) Shadow-removed

Fig. 7. Examples of ISTD.



(a) Longitudinal Crack (b) Transverse Crack (c) Net Crack

(d) Block Repair (e) Strip Repair (f) Pothole

Fig. 8. Examples of PDD.

2) Pavement Damage Dataset: The Pavement Damage Dataset (PDD) used in this paper is derived from a proprietary dataset collected using front-view video streams captured by vehicle-mounted equipment. Data collection was carried out on asphalt pavements with typical pavement damage across various highway sections in cities such as Jiujiang (Jiangxi Province), Linyi (Shandong Province), and Guiyang (Guizhou Province) in China. The video streams have a resolution of 1920×1080 and a frame rate of 30 frames per second.

PDD was annotated frame-by-frame using the open-source software DarkLabel. The dataset includes six categories of pavement damage: transverse cracks, longitudinal cracks, net cracks, potholes, block repairs, and strip repairs. The annotation process defines the first frame where damage appears and the last frame where it disappears as a complete labeling process. Bounding boxes are used to label the damage in the first and last frames, while intermediate frames are automatically annotated using interpolation, completing the damage annotation for the sequence.

A systematic data cleaning process was conducted on the constructed PDD to reduce issues such as poor annotation quality, sample imbalance, and noise interference. After data cleaning, the images were split into training and test sets in a 9:1 ratio. The specific number of images in each set is shown in Table II.

B. Settings

The hardware utilized for training and testing in this study includes an Intel(R) Core(TM) i7-12800H CPU @ GHz and an

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TABLE II
NUMBER OF EACH TYPE OF DAMAGE IN PDD

Types of damage	Number of Dataset		
	Total	Training Set	Test Set
Longitudinal Crack	14656	13190	1465
Transverse Crack	14788	13309	1479
Net Crack	3092	2783	309
Pothole	2272	2044	228
Block Repair	15584	14026	1558
Strip Repair	14084	12675	1409

NVIDIA GeForce GTX 3090 GPU. The software environment comprises Ubuntu 20.04 LTS, PyTorch 2.1.2, and CUDA 11.8.

For shadow removal, the number of epochs for RACycle-GAN is set to 200, with the Adam optimizer of an initial learning rate of 0.0002, which gradually decreases to 0 from epoch 100. The hyperparameter λ which controls the importance of cycle consistency in the loss function, is set to 10 according to the default setting [59].

For damage detection, the number of epochs for DDE-YOLOv8 is set to 300, with a batch size of 64 and an AdamW optimizer with an initial learning rate of 0.01 and a momentum value of 0.9. The initial confidence threshold is set to 0.001 or 0.01, allowing the model to output as many bounding boxes as possible during training or inference, ensuring that potential targets are not missed. The non-maximum suppression threshold is set to 0.6, which helps reduce redundant detections while maintaining detection accuracy.

C. Evaluation Metrics

1) Shadow Removal Evaluation Metrics: This study employs the Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and Root Mean Square Error (RMSE) to assess and compare the performance of RACycle-GAN and other algorithms in shadow removal tasks.

(a) PSNR is a metric used to evaluate the quality of image generation. It is calculated based on the noise level between the original image and the generated image, with higher values indicating better generation quality. PSNR can be calculated using the formula:

$$PSNR = 10 \times \log_{10} \frac{MAX_I^2}{(x-y)^2} \tag{9}$$

where MAX_I is the maximum possible pixel value of the image, x is the real shadow-free image and y is the generated shadow-removed image.

(b) SSIM is another metric that assesses the quality of images based on structural information. It takes changes in luminance, contrast, and structure into account, providing a more comprehensive measure of image similarity. SSIM is calculated as:

$$SSIM = mean\left(\sum \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}\right) \tag{10}$$

where μ_x and μ_y represent the variables for measuring luminance, while σ_x^2 and σ_y^2 assess contrast, and $\sigma_x y$ quantifies structural similarity. Constants C_1 and C_2 are included to avoid division by zero.

(c) RMSE quantifies the difference between the original and reconstructed images by taking the square root of the average of the squared differences. It is defined as:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x-y)^2} \tag{11}$$

2) Damage Detection Evaluation Metrics: For damage detection, the primary evaluation metrics include Accuracy, Precision, Recall, Mean Average Precision (mAP), and F1 Score to assess the performance of the algorithm.

The confusion matrix for these scenarios is summarized in Table III, providing a comprehensive overview of the performance.

TABLE III
CONFUSION MATRIX

Label	Prediction	
	Predict as Positive	Predict as Negative
True	True Positive (TP)	False Negative (FN)
False	False Positive (FP)	True Negative (TN)

(a) Accuracy (Acc) typically refers to classification accuracy in object detection, which is a measure of how correctly the model identifies the class labels of detected objects. Acc is calculated as below:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{12}$$

(b) Precision (P) measures the proportion of correctly predicted positive samples among all predicted positive samples for a particular class. It is defined as:

$$Precision = \frac{TP}{TP + FP} \tag{13}$$

(c) Recall (R) measures the proportion of correctly predicted positive samples among all actual positive samples for a particular class. It is defined as:

$$Recall = \frac{TP}{TP + FN} \tag{14}$$

(d) F1-score is the harmonic mean of Precision and Recall, providing a single measure that balances both metrics. The formula for calculating the F1 score is as follows:

$$F_{1k} = \frac{2 \cdot Precision_k \cdot Recall_k}{Precision_k + Recall_k} \tag{15}$$

To obtain the final result, the F1 Scores for each class can be averaged. The average F1 Score can be calculated using the following formula:

$$F_1 = \frac{1}{n}\sum_{i=1}^{k} F_{1i} \tag{16}$$

(e) Intersection over Union (IoU) is a metric used in object detection tasks to measure the overlap between the predicted bounding box (BBox) and the ground truth (actual) bounding box, which is a key metric in determining how well the model

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Fig. 9. Shadow removal results of RACycle-GAN.

is performing in terms of object localization. IoU is calculated as:

$$IoU = \frac{area(BBox_{pred} \cap BBox_{GT})}{area(BBox_{pred} \cup BBox_{GT}} \tag{17}$$

(f) Mean Average Precision (mAP) measures the overall performance of a model by calculating the average precision (AP) for each class and then averaging these values across all classes.

$$mAP = \frac{1}{n}\sum_{i=1}^{n} AP(i) \tag{18}$$

(g) mAP@50 and mAP@95 are specific forms of mAP at IoU = 0.50 and IoU = 0.95. mAP@50 is a relatively lenient metric because a 50% overlap is often sufficient for the detection to be counted as correct. mAP@95 refers to the Average Precision at IoU = 0.95. Because mAP@95 requires much more precise localization of the object, mAP@95 tends to result in lower scores compared to mAP@50.

D. Results of Shadow Removal

Figure 9 illustrates the shadow removal results of RACycle-GAN. The results demonstrate a significant improvement in shadow removal, effectively preserving the original pavement damage features.

E. Results of Damage Detection

Figure 10 illustrates the damage detection results of DDE-YOLOv8. The experimental results demonstrate that DDE-YOLOv8 accurately locates and classifies pavement damages, showing outstanding performance in identifying multi-scale damages.

As shown in Figure 10, most of the damages occupy a very small proportion in the images, and their shapes and textures are quite complex. Despite this, DDE-YOLOv8 is still able to accurately detect these damages, even in the case of very small potholes or subtle cracks.

After applying RACycle-GAN, shadows are effectively removed, thus mitigating shadow interference. The shadows on the road surface are transformed into pseudo-road surfaces that resemble the pavement, enhancing the subsequent ability of the damage detection model to withstand shadow interference in complex scenes.

TABLE IV

COMPARISON OF SHADOW REMOVAL ALGORITHMS ON ISTD

Metrics	Algorithms			
	Retinex	Mask-ShadowGAN	ST-CGAN	RACycle-GAN
PSNR ↑	17.82	28.89	27.73	**29.57**
SSIM ↑	0.74	0.82	0.90	**0.91**
RMSE ↓	8.36	7.61	7.47	**7.22**

V. DISCUSSION

A. Comparison of Shadow Removal Algorithms

1) Performance Comparison: To accurately assess the performance of RACycle-GAN, comparison experiments are conducted between several shadow removal methods, including Retinex [56] based on physical image processing, Mask-ShadowGAN [20], and ST-CGAN [44]. This section employs image quality evaluation metrics, including PSNR, SSIM, and RMSE to quantitatively evaluate the shadow removal performance of the different algorithms. The comparison of shadow removal experiments on ISTD is summarized in Table IV, and some results are illustrated in Figure 11. The experimental results for the compared algorithms are derived from reproduction experiments from their respective source codes.

The proposed RACycle-GAN demonstrates superior performance in shadow removal. Compared to Retinex, Mask-ShadowGAN, and ST-CGAN, the image quality metrics SSIM are improved by 0.17, 0.09, and 0.01, respectively, while the PSNR values are enhanced by 11.75, 0.58, and 1.84, respectively. The highest PSNR and SSIM values for RACycle-GAN indicate that the quality of the images after shadow removal is significantly improved, and the SSIM between shadow-removed images and shadow-free images is high. Additionally, the lowest RMSE value suggests minimal image error, achieving effective shadow removal while reducing image distortion. In addition to the metric improvements, RACycle-GAN has additional advantages that can be highlighted. Compared to Mask-ShadowGAN which relies on shadow masks to effectively learn the mapping between shadow and shadow-free domains, RACycle-GAN achieves superior performance without requiring shadow masks. ST-CGAN consists of two separate GAN networks: one for shadow detection and the other for shadow removal. Compared to ST-CGAN, RACycle-GAN not only requires no shadow masks but also has a simpler network architecture and achieves faster computational speed.

As shown in Figure 11, shadow removal in Mask-ShadowGAN and ST-CGAN presents several issues.

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Fig. 10. Damage detection results of DDE-YOLOv8.

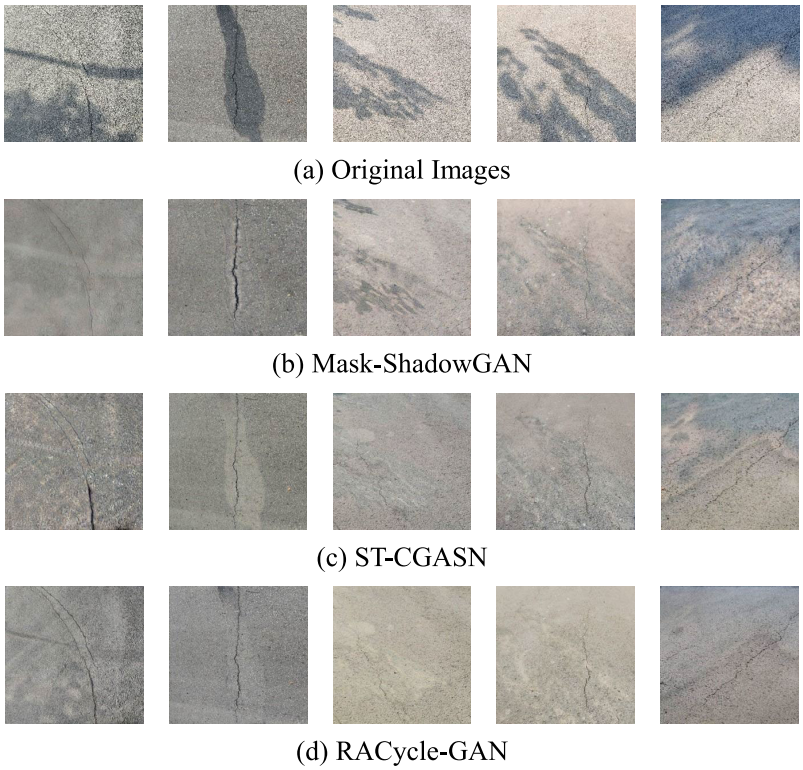

Fig. 11. Shadow removal results of different algorithms.

First, residual shadows remain after removal, such as shadow removal by Mask-ShadowGAN in the third, fourth, and fifth original images, as well as shadow removal by ST-CGAN in the fifth original image. Second, shadow removal inadvertently affects the texture of the damages, as observed in shadow removal by Mask-ShadowGAN in the first and fourth original

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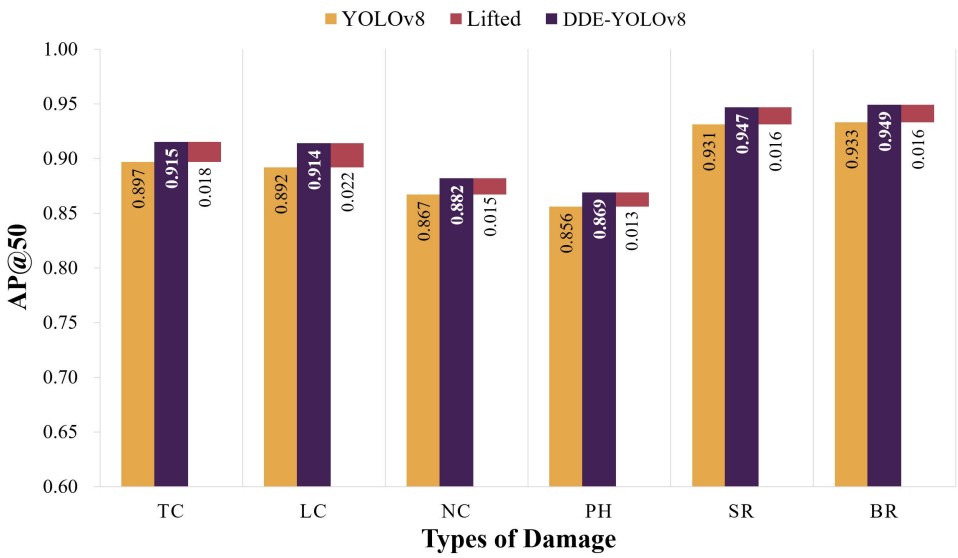

Fig. 12. AP@50 and mAP@50 comparison of DDE-YOLOv8 and the original YOLOv8.

TABLE V
ABLATION EXPERIMENTS OF RACYCLE-GAN ON ISTD (REDB: RESIDUAL ENCODER-DECODER BLOCK, AM: ATTENTION MECHANISM, CC: CYCLE CONSISTENCY)

REDB	AM	CC	RMSE	PSNR	SSIM
-	-	-	10.36	17.84	0.51
✓	-	-	8.11	22.19	0.85
-	✓	-	9.72	25.34	0.88
-	-	✓	7.98	24.47	0.74
✓	✓	✓	**7.22**	**29.57**	**0.91**

images, and in shadow removal by ST-CGAN in the second original image.

2) Ablation Experiment: To quantify the effectiveness of the three components (Residual Encoder-Decoder Block, attention mechanisms, and cycle consistency) on the performance enhancement of RACycle-GAN, we conduct ablation experiments on ISTD to analyze their impact on the shadow removal algorithm.

Table V demonstrate that all three components have a positive effect on performance improvement. Above all, the combination of three components gives RACycle-GAN the most outstanding performance. The use of the Residual Encoder-Decoder Block maintains the flow of information in deep networks, effectively adjusting the shadow regions of the image while preserving the original image information. The incorporation of attention mechanisms guides the network to focus more on regions that exhibit shadow features. Introducing cycle consistency prevents mode collapse, enabling the network to utilize relevant information more efficiently.

B. Comparison of Damage Detection Algorithms

1) Performance Comparison: To validate the performance enhancement of DDE-YOLOv8 relative to the original version, mAP@50 for all types of damage and AP@50 for each type are selected, shown in Figure 12.

Comparing the mAP of DDE-YOLOv8 and the original YOLOv8 on the test set, DDE-YOLOv8 shows improvements in all damage types, increasing by 0.016, 0.018, 0.050, 0.024, 0.022 and 0.020, respectively, which illustrates the excellent performance enhancement of DDE-YOLOv8.

The specific detection results are illustrated in Figure 13. As illustrated in the figure, the original YOLOv8 is prone to missing detections when facing small damages and simple textures (such as transverse and longitudinal cracks). Additionally, under the interference of pavement shadows, the original YOLOv8 misses the detection, while DDE-YOLOv8 successfully detects the damage. These experimental comparisons demonstrate that DDE-YOLOv8 is better at perceiving complex damages, exhibiting a higher detection performance.

To better assess the effectiveness of DDE-YOLOv8, comparisons are made among the five object detection algorithms: Faster R-CNN [36], Cascade R-CNN [5], SSD [27], the original YOLOv8, and DDE-YOLOv8 based on metrics Acc, P, R, F1-Score, mAP@50, mAP@95 and IoU. The comparative results are presented in Table VI.

From the data in the table, it is evident that DDE-YOLOv8 has a significant advantage in terms of detection performance. Compared to Faster R-CNN, Cascade R-CNN, SSD, and the original YOLOv8, Acc is enhanced by 5.1%, 7.2%, 11.7%, 2.6%, respectively. Specifically, the IoU surpasses other models by 3.18%, 6.66%, 15.86%, and 0.73%, indicating that DDE-YOLOv8 has outstanding performance on both classification and localization.

2) Ablation Experiment: To quantify the effectiveness of the two components (deformable convolution DCNv3 and CBAM) on the performance enhancement of the original YOLOv8, this paper conducts an ablation experiment on the PDD dataset.

Based on the data in the table, Both DCNv3 and CBAM enhance the performance of the original YOLOv8. Replacing the 1×1 convolution with a 3×3 deformable convolution allows the model to better understand the boundary information of smaller objects, effectively integrating shallow object information into higher-level feature maps and improving detection accuracy. Additionally, the CBAM module enhances

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(a) The Original YOLOv8 (b) DDE-YOLOv8 (c) The Original YOLOv8 (d) DDE-YOLOv8

Fig. 13. Damage Detection Results of DDE-YOLOv8 and The Original YOLOv8.

TABLE VI
COMPARISON OF DAMAGE DETECTION ALGORITHMS ON PDD

Algorithm	Acc	P	R	F1-Score	mAP@50	mAP@95	IoU
Faster R-CNN	0.872	0.858	0.867	0.863	0.865	0.699	0.805
Cascade R-CNN	0.851	0.864	0.872	0.868	0.859	0.702	0.770
SSD	0.806	0.831	0.861	0.846	0.841	0.674	0.678
Original YOLOv8	0.897	0.865	0.887	0.876	0.892	0.710	0.830
DDE-YOLOv8	**0.923**	**0.909**	**0.914**	**0.913**	**0.908**	**0.715**	**0.837**

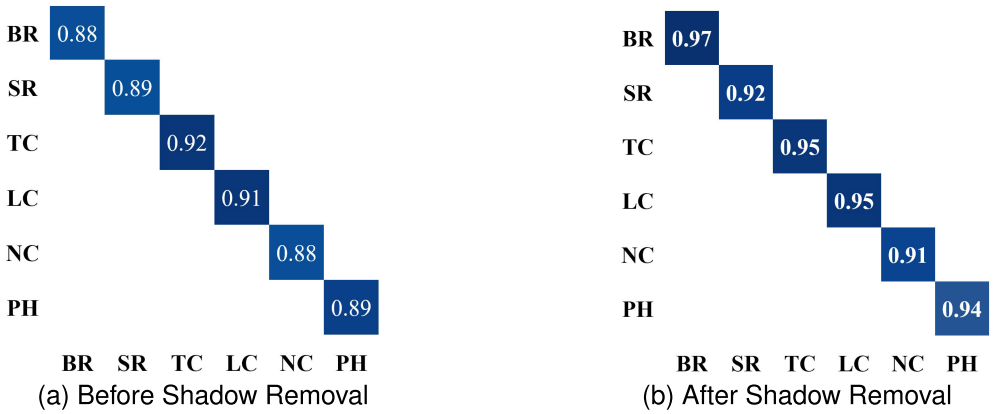

Fig. 14. Detection accuracy comparison of the original YOLOv8 on different datasets (TC: Transverse Crack, LC: Longitudinal Crack, NC: Net Crack, PH: Pothole, SR: Strip Repair, BR: Block Repair).

detection performance, likely due to its ability to adaptively strengthen the awareness of the model to key feature representation through CBAM, thereby improving the model's capacity for extracting damage features.

C. Detection Performance Enhancement With RACycle-GAN

In order to explore the impact of RACycle-GAN on the performance of pavement damage detection algorithms, some experiments are conducted in this section. First, PDD is split into training and test sets in a 10:1 ratio, denoted as D^s_{train} and D^s_{test}. Employ the RACycle-GAN on D^s_{train} and D^s_{test} respectively, then achieve shadow-removed training dataset D^{sr}_{train} and shadow-removed test dataset D^{sr}_{test}. The performance gap resulting from RACycle-GAN is quantified by training and testing algorithms with the shadowed dataset and the shadow-removed dataset.

The confusion matrix in Figure 14 provides a clear result of the detection accuracy of different types of damage. After

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TABLE VII
ABLATION EXPERIMENTS OF DDE-YOLOv8 ON PDD

DCNv3	CBAM	Acc	P	R	F1-Score	mAP@50	mAP@95	IoU
-	-	0.897	0.865	0.887	0.876	0.892	0.710	0.830
✓	-	0.918	0.883	0.895	0.879	0.918	0.711	0.827
-	✓	0.912	0.874	0.903	0.885	0.901	0.714	0.836
✓	✓	**0.923**	**0.909**	**0.914**	**0.913**	**0.908**	**0.715**	**0.837**

TABLE VIII
COMPARISON OF DAMAGE DETECTION ALGORITHMS ON D^s_{test} AND D^{sr}_{test}

Dataset	Algorithm	Acc	P	R	F1-Score	mAP@50	mAP@95	IoU
D^s	Faster R-CNN	0.872	0.858	0.867	0.863	0.865	0.699	0.805
	Cascade R-CNN	0.851	0.864	0.872	0.868	0.859	0.702	0.770
	SSD	0.806	0.831	0.861	0.846	0.841	0.674	0.678
	Original YOLOv8	0.897	0.865	0.887	0.876	0.892	0.710	0.830
	DDE-YOLOv8	0.923	0.909	0.914	0.913	0.908	0.715	0.837
D^{sr}	Faster R-CNN	0.880	0.865	0.872	0.868	0.894	0.724	0.823
	Cascade R-CNN	0.862	0.873	0.875	0.874	0.887	0.711	0.794
	SSD	0.829	0.862	0.879	0.870	0.869	0.698	0.689
	Original YOLOv8	0.907	0.876	0.892	0.884	0.903	0.719	0.841
	DDE-YOLOv8	**0.934**	**0.921**	**0.936**	**0.928**	**0.927**	**0.734**	**0.863**

applying RACycle-GAN, the performance of the original YOLOv8 shows a noticeable improvement. Specifically, the accuracy of the six types of damages increased by 9%, 3%, 3%, 4%, 3%, and 5%, respectively. The effective removal of shadows aids in reducing misjudgments by the model in damage detection, thereby enhancing overall prediction performance.

Furthermore, the data presented in Table VIII strongly demonstrates that after shadow removal using RACycle-GAN, all tested damage detection algorithms exhibited a certain degree of performance improvement, which proves that RACycle-GAN performs exceptionally well in the task of shadow removal, making a significant contribution to enhancing pavement damage detection. Most importantly, the combination of RACycle-GAN and DDE-YOLOv8 achieves the best performance in pavement damage detection, demonstrating the effectiveness of the shadow-robust damage detection method in this paper.

VI. CONCLUSION

Current pavement damage detection algorithms are often interfered with by pavement shadows and face challenges in detecting complex, multi-scale damages. Aiming at solving the problems above, we propose a shadow-robust pavement damage detection framework by combining RACycle-GAN and DDE-YOLOv8. Through extensive experiments, the superior performance of shadow removal based on RACycle-GAN and the damage detection of DDE-YOLOv8 has been fully demonstrated. By combining RACycle-GAN for shadow removal and DDE-YOLOv8 for pavement damage detection, significant advantages are observed compared to other methods. This approach effectively performs in accomplishing the task of pavement damage detection.

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